

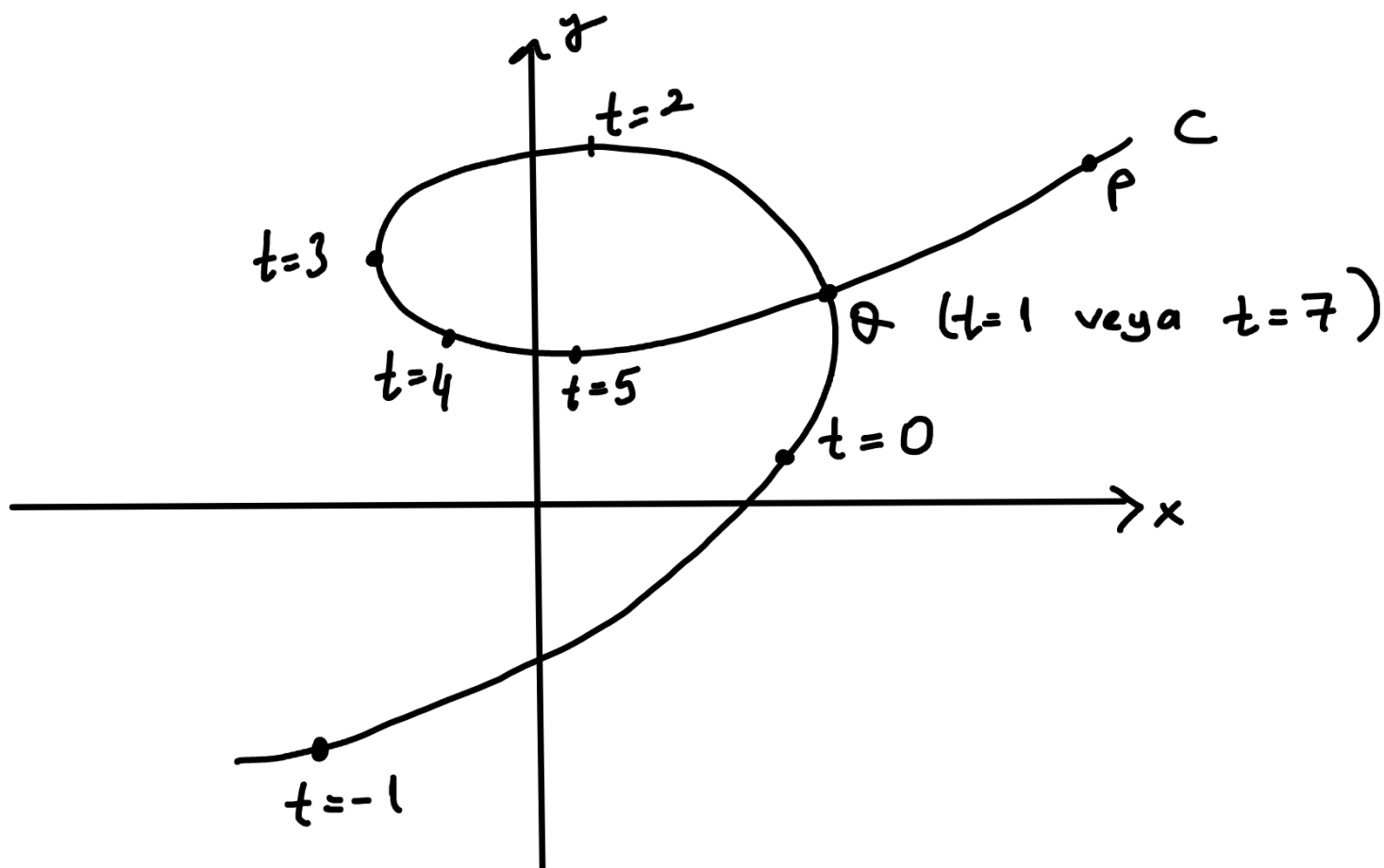
8.3. Parametrik Formda Eğriler

Tanım. t bir parametre $t \in I$

$$x = x(t), \quad y = y(t)$$

$$(x(t), y(t))$$

$$C : \quad x = x(t), \quad y = y(t), \quad t \in I$$



$$x(1) = x(7)$$

$$y(1) = y(7)$$

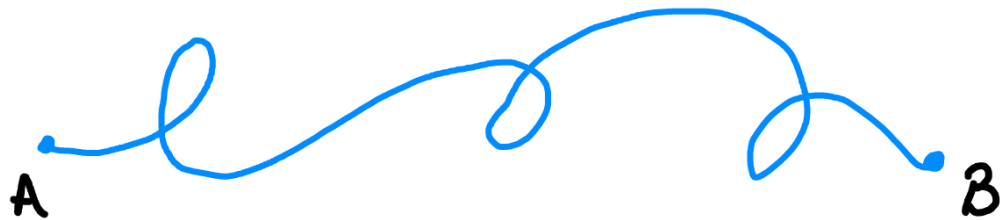
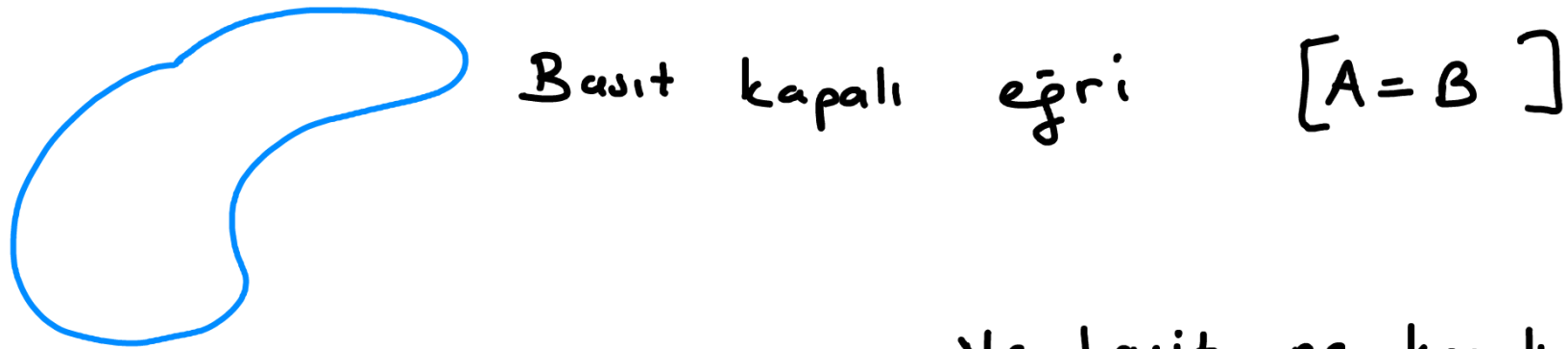
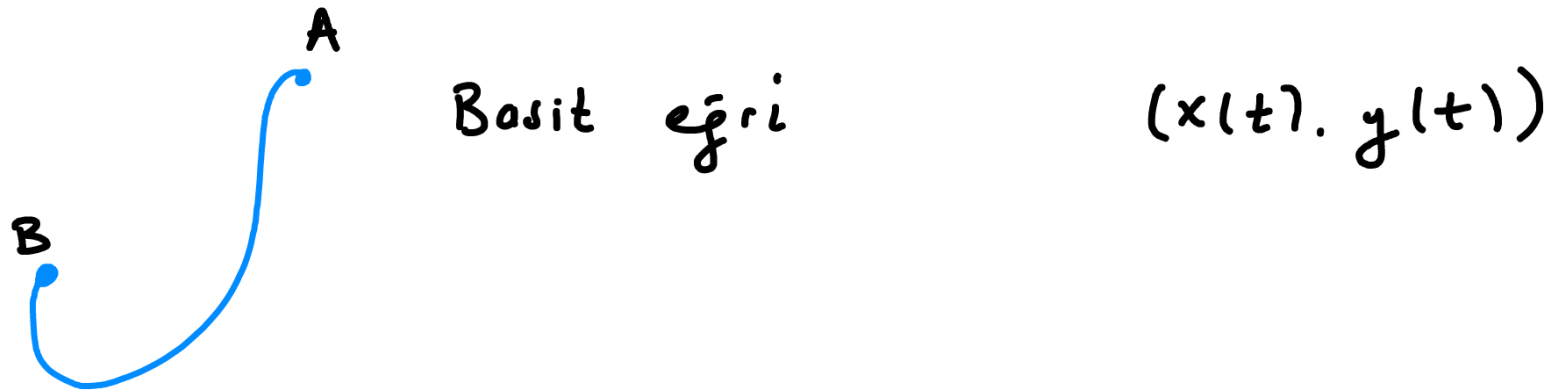
Drnek1. $y = f(x) \quad ; \quad a \leq x \leq b$

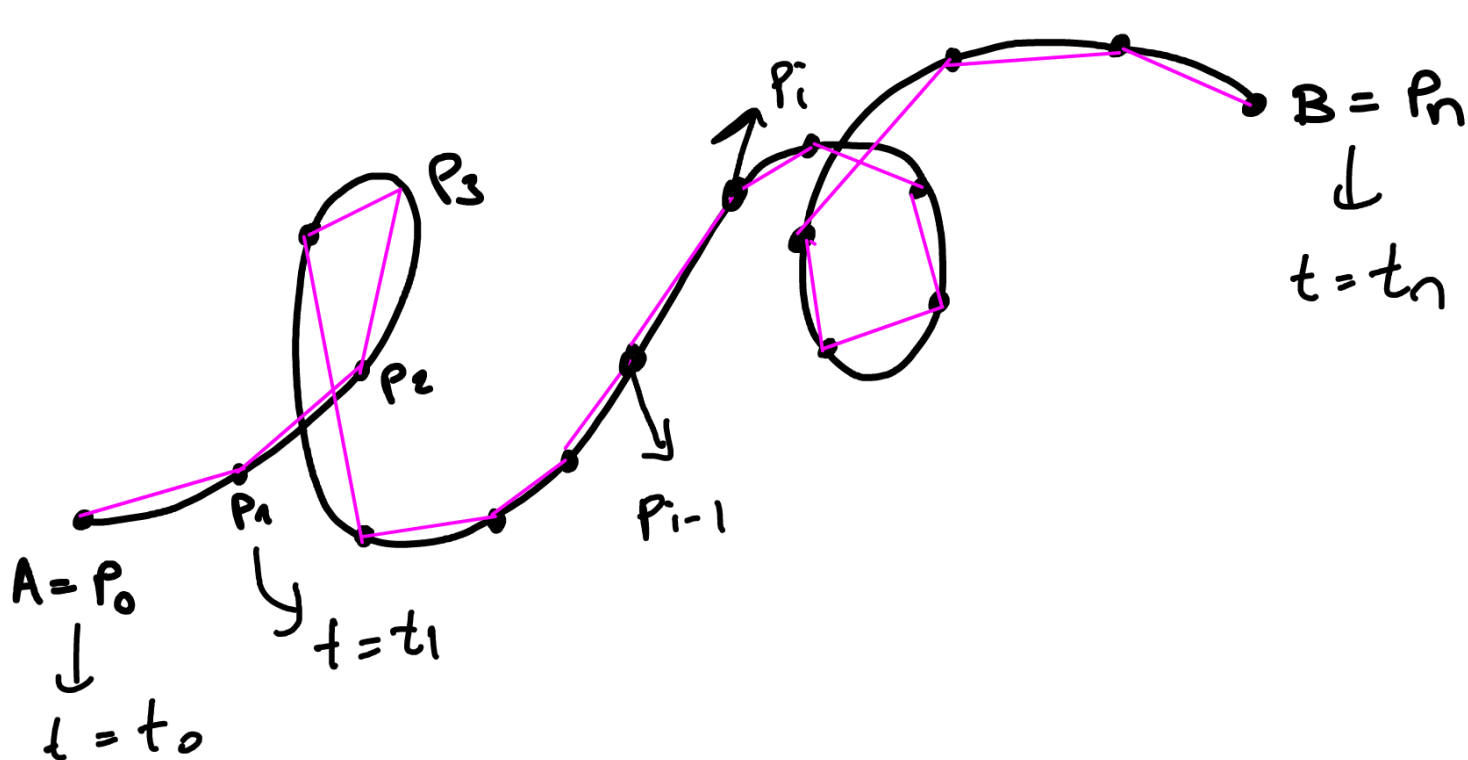
$$x = t \quad ; \quad y = f(t) \quad a \leq t \leq b$$

Veya $x = g(y) \quad a \leq y \leq b$

$$y = t \quad , \quad x = g(t) \quad ; \quad a \leq t \leq b$$

8.4 Parametrik Eğrilerde Uzunluk





$$(x(t), y(t))$$

$$A = (x(a), y(a))$$

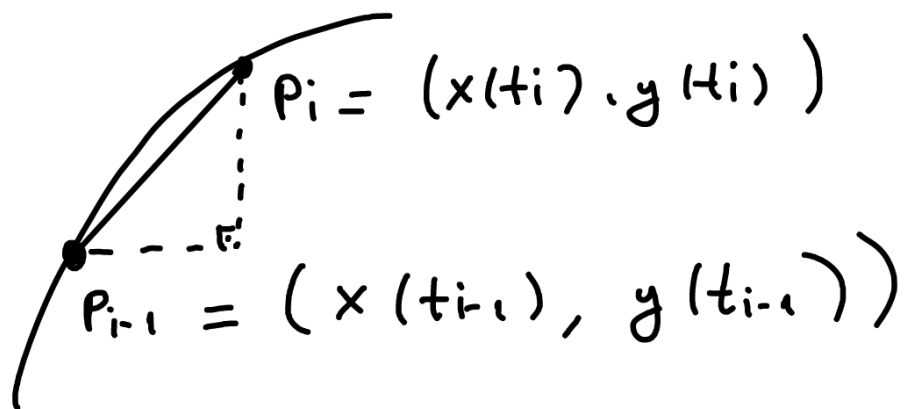
$$B = (x(b), y(b))$$

$$a = t_0 < t_1 < \dots < t_{n-1} < t_n = b$$

$$L \approx \sum_{i=1}^n |P_{i-1} P_i|$$

$$\Delta t_i = t_i - t_{i-1}, \quad M = \max\{\Delta t_1, \Delta t_2, \dots, \Delta t_n\}$$

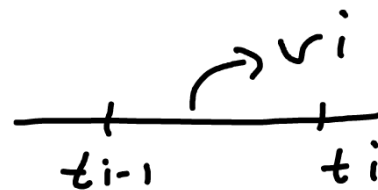
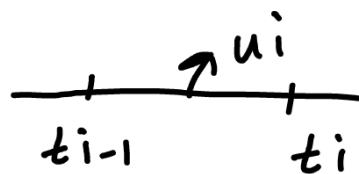
$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n |P_{i-1} P_i|$$



$$f(x) - f(y) = f'(z) \cdot (x - y)$$

$$|P_{i-1} P_i| = \left[(x(t_i) - x(t_{i-1}))^2 + (y(t_i) - y(t_{i-1}))^2 \right]^{1/2}$$

$$= [x'(u_i)]^2 (\Delta t_i)^2 + [y'(v_i)]^2 (\Delta t_i)^2$$



$$\sum_{i=1}^n |P_{i-1}, P_i| = \sum_{i=1}^n \sqrt{x'(u_i)^2 + y'(v_i)^2} \Delta t_i$$



$$\sqrt{x'(t)^2 + y'(t)^2}$$

$$L = \int_a^b \sqrt{x'(t)^2 + y'(t)^2} dt$$

örnek. $x = a(t - \sin t)$, $y = a(1 - \cos t)$; $0 \leq t \leq 2\pi$, $a > 0$

$$x'(t) = a(1 - \cos t)$$

$$y'(t) = a \sin t$$

$$\begin{aligned} [x'(t)]^2 + [y'(t)]^2 &= a^2 - 2a^2 \cos t + a^2 \cos^2 t + a^2 \sin^2 t \\ &= a^2 (2 - 2 \cos t) \end{aligned}$$

$$= 2a^2 \left(1 - \cos^2 \frac{t}{2} + \sin^2 \frac{t}{2} \right)$$

$$= 4a^2 \sin^2 \frac{t}{2}$$

$$L = \int_0^{2\pi} \sqrt{4a^2 \sin^2 \frac{t}{2}} dt = 8a$$

Not. C sürekli türevlenebilir $y = f(x)$, $a \leq x \leq b$
fonksiyonunun grafiği olsun.

O zaman C aşağıdaki parametrik gösterime
sahip olur:

$$x = t, \quad y = f(t) \quad ; \quad a \leq t \leq b \Rightarrow x'(t) = 1, \quad y'(t) = f'(t)$$

Eğrinin uzunluğu

$$L = \int_a^b \sqrt{1 + (f'(t))^2} \, dt = \int_a^b \sqrt{1 + (f'(x))^2} \, dx$$

veya

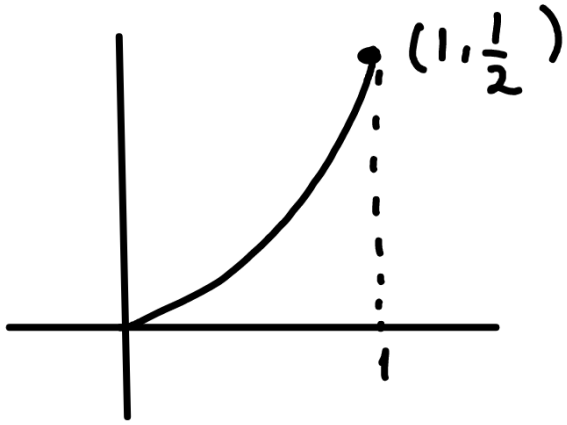
Örnek. $C: y = \sqrt{a^2 - x^2}$, $-a \leq x \leq a$ yarı çemberi

olsun:

$$\frac{dy}{dx} = -\frac{x}{\sqrt{a^2 - x^2}}$$

$$L = \int_{-a}^a \sqrt{1 + \frac{x^2}{a^2 - x^2}} dx = \pi a$$

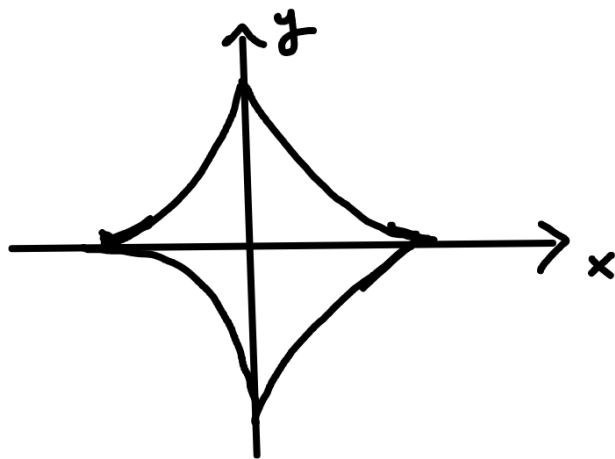
örnek. $y = \frac{1}{2}x^2$ eğriinin orijinden $(1, \frac{1}{2})$ noktasına kadar olan uzunluğunu bulunuz.



$$L = \int_0^1 \sqrt{1 + x^2} \, dx$$

dr nek.

$$x = a \cos^3 t, \quad y = a \sin^3 t; \quad 0 \leq t \leq 2\pi, \quad a > 0$$



$$L = 4 \int_0^{\pi/2} \sqrt{x'(t)^2 + y'(t)^2} dt$$

$$\begin{aligned} x'(t)^2 + y'(t)^2 &= 9a^2 (\cos^4 t \sin^2 t + \sin^4 t \cos^2 t) \\ &= 9a^2 \cos^2 t \sin^2 t (\cos^2 t + \sin^2 t) \end{aligned}$$

$$L = 12a \int_0^{\pi/2} \cos t \sin t dt = 6a \int_0^{\pi/2} \sin 2t dt = 6a$$

Örnek.

$$X = \frac{1}{6} y^3 + \frac{1}{2y}, \quad 1 \leq y \leq 3$$

eğrisinin uzunluğunu

bulunuz.

Çözüm.

$$L = \int_1^3 \sqrt{1 + (g'(y))^2} dy$$

$$g'(y) = \frac{y^2}{2} - \frac{1}{2y^2}$$

$$L = \int_1^3 \sqrt{1 + \frac{y^4}{4} + \frac{1}{4y^4} - \frac{1}{2}} dy$$

$$= \int_1^3 \sqrt{\frac{1}{2} + \frac{y^4}{4} + \frac{1}{4y^4}} dy = \int_1^3 \sqrt{\left(\frac{y^2}{2} + \frac{1}{2y^2}\right)^2} dy$$

$$= \frac{1}{2} \int_1^3 \left(y^2 + \frac{1}{y^2} \right) dy = \frac{1}{2} \left(\frac{y^3}{3} - \frac{1}{y} \right) \Big|_1^3 = 14/3$$

Örnek.

$y = \frac{1}{8} x^2 \ln x$, $1 \leq x \leq e$ eğri uzunluğunu bulunuz.

Çözüm.

$$f'(x) = \frac{x}{4} - \frac{1}{x}$$

$$L = \int_1^e \sqrt{1 + (f'(x))^2} dx = \int_1^e \sqrt{1 + \frac{x^2}{16} + \frac{1}{x^2} - \frac{1}{2}} dx$$

$$= \int_1^e \sqrt{\frac{x^2}{16} + \frac{1}{x^2} + \frac{1}{2}} dx = \int_1^e \sqrt{\left(\frac{x}{4} + \frac{1}{x}\right)^2} dx$$

$$= \int_1^e \left(\frac{x}{4} + \frac{1}{x}\right) dx = \frac{x^2}{8} + \ln x \Big|_1^e = \frac{1}{8} \cdot (e^2 + 7)$$